



**Baker
College**

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IS EARNED HERE.

Micro-Level PLC Programming

Programmable Logic Controllers

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Agenda

Digital vs. Analog

Analog Types

Analog Devices

PID Control

Student Learning Outcomes

1. Create programs in Ladder Logic using the following instructions in an application
2. Program PLC timers and counters

Digital vs. Analog

Digital Defined

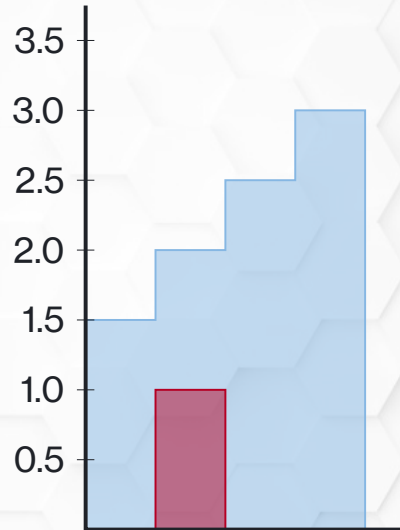
- Data or signals that have a fixed number of discrete values within their range
- Devices that use such data/signals



"Digital clock" by MichalPL, CC BY-SA 4.0, via [Wikimedia Commons](#)

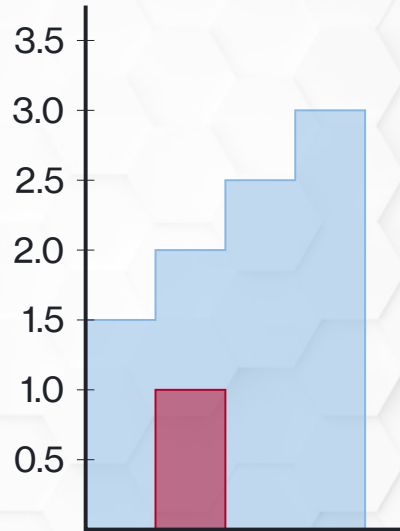
Digital Signals (1 of 2)

- *Binary* is a special case of digital—the range is 0-1
- All binary signals are digital, not all digital signals are binary



Digital Signals (2 of 2)

- Digital signals *can* have decimal points, even a *lot* of decimal points
- But they can still only be a fixed number of values
- For example this signal can be 2 or 2.5, nothing between



Analog Defined

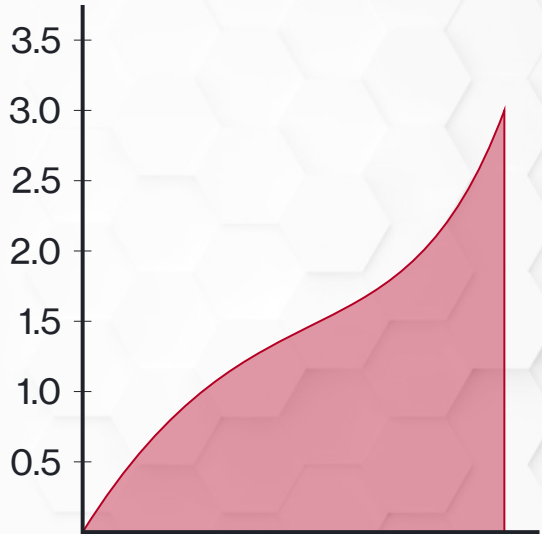
- Data or signals that vary continuously over a range of values
- Devices that use this type of data/signal



"Mercury-in-glass thermometer" by Anonimski, CC BY-SA 3.0, via [Wikimedia Commons](#)

Analog Signals

- Digital signals *can* have decimal points, even a *lot* of decimal points
- But they can still only be a fixed number of values
- For example this signal can be 2 or 2.5, nothing between



Why Analog?

- Motors use 69% of all electricity in the U.S. industrial sector¹
- Motors running at full speed use more energy than motors running at reduced speed

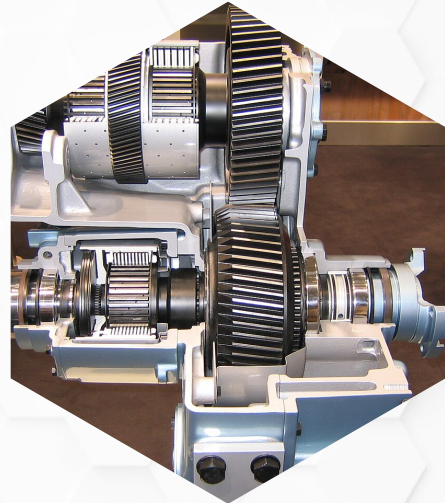


"Asynchronous electric motor" by Egzon123, CC BY-SA 3.0, via [Wikimedia Commons](#)

¹Department of Energy, "Industrial Electric Motor Market Assessment" (2020).

Controlling Motor Speed

- When a motor doesn't *need* to run at full speed you *could* use a gearbox...
- ...but that doesn't save any energy



"ZF gearbox at the Bauma trade fair 2004" by Aconcagua, [CC BY-SA 3.0](#), via [Wikimedia Commons](#)

Saving Motor Energy

- Save energy (\$) by reducing motor speed
- VFDs *always* have an analog input
- Most have 2: voltage and current
- Can accept either potentiometer or PLC analog output



"Mitsubishi frequency converter type D700" by Ignac.mihevc at Slovenian Wikipedia, [CC BY-SA 3.0](#), via [Wikimedia Commons](#)

Analog Types



Analog Devices

PID Control



Questions?

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